

Condensed Mathematics Meets Topological Quantum Field Theory: A Formally Verified Foundation

Abstract

We present a formally verified investigation into using condensed abelian groups (Clausen-Scholze) as a target category for topological quantum field theories, replacing topological vector spaces which fail to be abelian. Over six sessions of AI-assisted formal verification using Lean 4, Mathlib v4.28.0 [Mathlib], and Harmonic's Aristotle system, we establish:

1. **Symmetric monoidal structure on CondensedAb** (sorry-free, assembling existing Mathlib infrastructure via sheaf localization)
2. **Abstract TQFT transfer theorem** (sorry-free, braided monoidal functors preserve TQFT structure)
3. **Embedding functor SemiNormedGrp to CondensedAb** (sorry-free, first bridge between Mathlib's normed space and condensed libraries)
4. **Faithfulness of the embedding** (sorry-free) and a **disproof of fullness** for seminormed groups (counterexample: continuous additive maps need not be bounded without scalar structure)
5. **Commutative Frobenius algebras and a combinatorial 2d cobordism category** (sorry-free algebraic structure, sorry-free category definition)

The project comprises approximately 1280 active lines of Lean 4 across 5 files, with 3 remaining sorries (all with documented proof strategies) and 0 custom axioms. The accompanying formalization is available as a self-contained Lean 4 project buildable via `lake build`.

1. Introduction

1.1 The Categorical Obstruction in Infinite-Dimensional TQFT

A topological quantum field theory, in Atiyah's formulation [Atiyah1988], is a symmetric monoidal functor $Z : \text{Cob}(d+1)$ to $\mathcal{C}$ from a cobordism category to some symmetric monoidal target category $\mathcal{C}$. For finite-dimensional state spaces (Dijkgraaf-Witten, Reshetikhin-Turaev, and Turaev-Viro theories), the target $\mathcal{C} = \text{Vect}$ works perfectly.

The physically interesting theories force infinite-dimensional state spaces. Chern-Simons theory

[Witten1988] with non-compact gauge group $SL(2, \mathbb{C})$, quantum gravity partition functions, and path integrals over infinite-dimensional moduli all produce state spaces $Z(\Sigma)$ that cannot live in finite-dimensional Vect. The natural target becomes topological vector spaces.

Here a precise categorical obstruction arises. The category TopAb of topological abelian groups is **not abelian**: a continuous bijective homomorphism need not have a continuous inverse, so a morphism can be simultaneously monic and epic without being an isomorphism. This propagates through the homological algebra machinery. Exact sequences break, derived functors misbehave, and the TQFT gluing axiom (which decomposes state spaces via exact sequences when cutting manifolds) becomes ill-defined.

The completed tensor product compounds the problem. It is not exact: tensoring a short exact sequence with a topological vector space can destroy exactness. Since the TQFT tensor product (disjoint union of boundaries corresponds to tensor product of state spaces) requires this operation, infinite-dimensional TQFTs built on TopVect are categorically deficient.

1.2 The Condensed Fix

Clausen and Scholze’s condensed mathematics [Scholze2019] resolves this obstruction. The category of condensed abelian groups (sheaves of abelian groups on the coherent site of compact Hausdorff spaces) is an abelian category with all limits and colimits. Every morphism has a kernel and cokernel. Every monic epic is an isomorphism.

Liquid vector spaces, a refinement of condensed modules, admit an exact completed tensor product. This was the content of the Liquid Tensor Experiment [Scholze2021, LTE2022], Lean-verified by Commelin et al. in 2022. The exactness of this tensor product is precisely the property TopVect lacks and TQFT gluing requires.

1.3 This Paper

We report on a formal verification effort investigating whether CondensedAb can serve as a TQFT target category. The work was conducted iteratively over six sessions using Harmonic’s Aristotle system [Aristotle2025] for Lean 4 proof search, with Claude (Anthropic) providing mathematical synthesis, prompt engineering, and analysis.

The project produced both expected confirmations and genuine mathematical surprises. The monoidal structure on CondensedAb turned out to already exist in Mathlib (we just had to find it). The embedding from seminormed groups turned out not to be full (a counterexample was discovered during formalization). Both findings reshaped the project’s direction in real time.

1.4 Navigating the Mathematical Landscape

To contextualize where this work sits within modern mathematics, we developed MathFlowChart, an interactive dependency map of 221 mathematical concepts and 491 connections. The chart (available as a self-contained HTML/React application) visualizes the path from foundational structures (set theory, category theory) through the algebraic hierarchy, analysis tower, and

algebraic geometry to the Scholze program and TQFT.

The relevant path through the chart runs: Category Theory to Topos Theory to Sheaf Cohomology to Condensed Sets to Condensed Abelian Groups to Liquid Vector Spaces, with a parallel path from Algebraic Topology to Homotopy Theory to TQFT to Chern-Simons Theory. The intersection of these paths is the subject of this paper. The chart also tracks the Langlands program connections (geometric Langlands via Fargues-Scholze uses the same condensed infrastructure) and the stable/chromatic homotopy theory connections that bridge topology and number theory.

2. The Formalization: Six Sessions

2.1 Session 1: Abstract TQFT Framework (LiquidTQFT.lean)

The first session established the categorical scaffolding.

Definition (AbstractTQFT). A TQFT with source category S and target category C (both braided monoidal) is a braided monoidal functor $Z : S \rightarrow C$.

```
structure AbstractTQFT
  (S : Type*) [Category S] [MonoidalCategory S] [BraidedCategory S]
  (C : Type*) [Category C] [MonoidalCategory C] [BraidedCategory C] where
  Z : S  $\Rightarrow$  C
  monoidal : Z.Monoidal
  braided : Z.Braided
```

We work at the level of braided (not symmetric) monoidal functors. This is strictly more general than Atiyah's original symmetric formulation and covers non-symmetric examples relevant to 3-dimensional TQFTs (e.g., Reshetikhin-Turaev invariants [ReshetikhinTuraev1991] from modular tensor categories). Lurie's cobordism hypothesis [Lurie2009] provides the fully extended framework in which these structures are most naturally understood. Since every symmetric monoidal category is braided, the results apply a fortiori to Atiyah-style TQFTs.

Theorem 2.1 (Transfer, sorry-free). If T is a TQFT valued in C and $F : C \rightarrow D$ is a braided monoidal functor, then T composed with F is a TQFT valued in D .

```
def AbstractTQFT.transfer
  (T : AbstractTQFT S C)
  (F : C  $\Rightarrow$  D) [hF : F.Monoidal] [hFb : F.Braided] :
  AbstractTQFT S D where
  Z := T.Z  $\gg$  F
  monoidal := by letI := T.monoidal; infer_instance
  braided := by letI := T.monoidal; letI := T.braided; infer_instance
```

The proof is Lean's typeclass inference composing the monoidal and braided structures. The fact that `infer_instance` suffices means the categorical plumbing is genuinely present in Mathlib.

Additional results from this session (all sorry-free): `CondensedAb` is abelian (`inferInstance`), has kernels, cokernels, and finite (co)limits, and satisfies the balanced category property that $\text{mono} + \text{epi} = \text{iso}$ (the exact property `TopAb` lacks).

The session also recorded structural observations about gluing: in any abelian category, short exact sequences provide the decomposition structure needed for TQFT gluing. These take exactness as a hypothesis rather than proving it for the condensed tensor product specifically. The substantive analytic claim (that the liquid tensor product is exact) depends on the Liquid Tensor Experiment and is not verified in this codebase.

2.2 Session 2: Manual Monoidal Construction (`CondensedMonoidal.lean`)

The second session attempted to build `MonoidalCategory CondensedAb` directly. The tensor product of two condensed abelian groups `F` and `G` is defined as the sheafification of their pointwise tensor product as presheaves:

```
tensorObj F G = sheafify(F.val tensor G.val)
```

This yielded a complete `MonoidalCategoryStruct` (all data defined, sorry-free) and proofs of 5 out of 10 `MonoidalCategory` axioms. The remaining 5 (associator naturality, unitor naturality, pentagon, triangle) all reduced to exactly two "sheafification comparison" isomorphisms:

```
sheafify(P tensor Q) ≅ sheafify(sheafify(P).val tensor Q)
```

These express that sheafification is a monoidal functor. Three proof approaches were identified (monoidal reflective subcategories, tensor preserving local isomorphisms, direct universal property). This analysis was essential for the breakthrough in Session 3.

2.3 Session 3: The Localization Breakthrough (`MonoidalViaLocalization.lean`)

The third session discovered that `Mathlib.CategoryTheory.Sites.Monoidal [MatlibSitesMon]` already contained the complete infrastructure. The monoidal structure on `CondensedAb` assembles from four existing components:

1. **Presheaf monoidal structure** (`Mathlib.CategoryTheory.Monoidal.FunctorCategory`): pointwise tensor on `CompHausop` to `Ab`.
2. **Sheafification as localization** (`Mathlib.CategoryTheory.Sites.Localization`): `presheafToSheaf` is a localization functor for the class `J.W` of local isomorphisms.
3. **`W` is monoidal** (`Mathlib.CategoryTheory.Sites.Monoidal`): `J.W` is closed under tensoring. The

proof uses the internal hom (monoidal closed structure), not stalks: for sheaf H , $\text{Hom}(F \otimes G, H)$ is isomorphic to $\text{Hom}(G, [F, H])$ by the tensor-hom adjunction, and $[F, H]$ is a sheaf.

4. **Localized monoidal category** (`Mathlib.CategoryTheory.Localization.Monoidal.Basic`): general construction for monoidal structures on localized categories.

Theorem 2.2 (sorry-free). `CondensedAb` is a symmetric monoidal category.

```
instance condensedAb_W_isMonoidal :
  ((coherentTopology CompHaus).W
    (A := ModuleCat (ULift ℤ))).IsMonoidal :=
  GrothendieckTopology.W.monoidal

instance condensedAb_monoidalCategory :
  MonoidalCategory CondensedAb :=
  Sheaf.monoidalCategory _ _
```

Six lines. Zero sorries. The sheafification functor is additionally verified to be monoidal and braided (both by `inferInstance`).

The discovery that the internal hom argument (rather than stalks) powers the `W.IsMonoidal` proof is mathematically significant: it means the result holds for any closed braided target category, not just abelian categories with enough points. This matters because `CompHaus` with the coherent topology does not have enough points in the classical sense.

2.4 Session 4: The Banach Embedding (`BanachEmbedding.lean`)

The fourth session audited `Mathlib` for connections between normed spaces and condensed mathematics. The finding: **zero existing bridge**. The `Analysis.NormedSpace` and `Condensed` library components were completely disjoint, with no file importing both.

The session constructed the first such bridge: a functor from seminormed abelian groups to condensed abelian groups.

Definition. For a seminormed abelian group V , the presheaf `banachPresheaf V` sends S in `CompHaus` to $C(S, V)$, the abelian group of continuous maps from S to V , with precomposition as the functorial action.

Theorem 2.3 (sorry-free). `banachPresheaf V` is a sheaf for the coherent topology on `CompHaus`.

The proof transfers the sheaf condition from the Type-valued `yonedaPresheaf` (which `Mathlib` already knows is a sheaf) through the forgetful functor. The main technical obstacle was universe arithmetic: `CompHaus` has `Category.{0,1}` while `ModuleCat` has `Category.{1,2}`, requiring `hasLimitsOfSizeShrink` and `preservesLimitsOfSize_of_univLE` to bridge.

Theorem 2.4 (sorry-free). There exists a functor `semiNormedGrpToCondensedAb` :
`SemiNormedGrp` to `CondensedAb`.

2.5 Session 5: Properties and a Counterexample (BanachEmbedding.lean, continued)

Theorem 2.5 (sorry-free). The embedding is faithful.

Proof: for any v in V , the constant function S to V at v is continuous. If two bounded homomorphisms agree on all $C(S,V)$, evaluating at the constant function at PUnit gives agreement on all elements of V .

Theorem 2.6 (Counterexample). The embedding `SemiNormedGrp` to `CondensedAb` is NOT full.

This was the project's most significant mathematical finding. The counterexample: let $V = c_0(\mathbb{N}, \mathbb{Z})$ (eventually-zero integer sequences with sup norm) and $W = \mathbb{Z}$. The summation map $f(a) = \sum(a_n)$ is additive and continuous (since $\|a\| < 1$ implies all $a_n = 0$ for integer sequences), but not bounded ($\|e_1 + \dots + e_n\| = 1$ while $f(e_1 + \dots + e_n) = n$). Postcomposition by f defines a valid natural transformation in `CondensedAb` with no lift to `SemiNormedGrp`.

Root cause: continuous additive maps between seminormed abelian groups need not be bounded. The standard proof that "continuous linear implies bounded" requires scalar multiplication by a dense subfield ($\mathbb{R}$, $\mathbb{C}$, or a nontrivially normed field). For bare abelian groups with a norm, no such scalar action exists.

Corrected scope: fullness holds when restricted to `NormedSpace R` for R a nontrivially normed field. The fully faithful embedding requires this restriction.

The session also constructed `SemiNormedGrp.hasFiniteProducts` (sorry-free, not previously in Mathlib), including the product fan with explicit operator norm bounds via `Finset.sup`.

2.6 Session 6: Combinatorial Cobordisms (Cob2.lean)

The sixth session formalized the algebraic structure classifying 2-dimensional TQFTs [Kock2003].

Definition (sorry-free). A `FrobeniusObj` in a monoidal category extends Mathlib's `MonObj` (monoid object) and `ComonObj` (comonoid object) with the Frobenius relation:

$$(\mu \text{ tensor } \text{id}) \text{ compose } (\text{id} \text{ tensor } \delta) = \delta \text{ compose } \mu$$

A `CommFrobeniusObj` additionally requires commutativity: $\mu \text{ compose } \beta = \mu$.

Theorem 2.7 (sorry-free). In any braided monoidal category, the unit object carries a canonical commutative Frobenius algebra structure. All 9 axioms are proved using Lean's `coherence` tactic (for monoidal coherence diagrams) and two braiding lemmas.

The 2-dimensional cobordism category Cob2 is defined as a category on the natural numbers (representing n disjoint circles), with morphisms generated by:

- μ : 2 to 1 (pants: two circles merge)
- η : 0 to 1 (cap: circle from nothing)
- δ : 1 to 2 (copants: circle splits)
- ϵ : 1 to 0 (cup: circle caps off)

plus composition, tensor product, and swap, quotiented by the Frobenius algebra relations.

3. Results Summary

3.1 Complete Theorem Inventory

Theorem	File	Status
CondensedAb is abelian	LiquidTQFT.lean	Sorry-free
Mono + epi = iso in abelian categories	LiquidTQFT.lean	Sorry-free
MonoidalCategory CondensedAb	LiquidTQFT.lean	Sorry-free
BraidedCategory CondensedAb	LiquidTQFT.lean	Sorry-free
SymmetricCategory CondensedAb	LiquidTQFT.lean	Sorry-free
AbstractTQFT.transfer	LiquidTQFT.lean	Sorry-free
W.IsMonoidal for coherent topology	MonoidalViaLocalization.lean	Sorry-free
presheafToSheaf is monoidal	MonoidalViaLocalization.lean	Sorry-free
presheafToSheaf is braided	MonoidalViaLocalization.lean	Sorry-free
banachPresheaf V is a sheaf	BanachEmbedding.lean	Sorry-free
semiNormedGrpToCondensedAb functor	BanachEmbedding.lean	Sorry-free
Embedding is faithful	BanachEmbedding.lean	Sorry-free
Embedding is full	BanachEmbedding.lean	Disproved
SemiNormedGrp.hasFiniteProducts	BanachEmbedding.lean	Sorry-free
FrobeniusObj, CommFrobeniusObj	Cob2.lean	Sorry-free

Trivial Frobenius on unit object	Cob2.lean	Sorry-free
Cob2 category definition	Cob2.lean	Sorry-free
TQFT2d.transfer	Cob2.lean	Sorry-free

3.2 Remaining Sorries

Sorry	File	Difficulty	Strategy
PreservesEqualizers (embedding)	BanachEmbedding.lean	Hard	4-5 layers of limit transfer via sheafToPresheaf
PreservesFiniteProducts (embedding)	BanachEmbedding.lean	Hard	ContinuousMap.piEquiv path
CommFrobeniusData.toCob2Functor	Cob2.lean	Tedious	Case analysis on quotient morphism type

3.3 Key Findings

Finding 1: CondensedAb monoidal structure already in Mathlib. The iterative process (axiom in Session 1, manual construction in Session 2, localization discovery in Session 3) demonstrates both the depth of the Mathlib library and the difficulty of discovering what it contains.

Finding 2: No prior bridge between normed spaces and condensed math. The embedding functor constructed in Session 4 is the first connection between Mathlib's `Analysis.NormedSpace` and `Condensed` components.

Finding 3: Fullness fails for seminormed groups. The condensed category sees strictly more morphisms than the normed category. This is not a formalization artifact but a genuine mathematical phenomenon: continuous additive maps without scalar structure need not be bounded.

Finding 4: SemiNormedGrp finite products not previously in Mathlib. The construction of `hasFiniteProducts` with explicit operator norm bounds is a small independent contribution.

Finding 5: Frobenius algebras formalize cleanly from existing Mon_/Comon_ infrastructure. The classification “2d TQFT = commutative Frobenius algebra” requires only extending Mathlib’s existing monoid and comonoid object typeclasses.

4. The MathFlowChart

To contextualize the mathematical dependencies involved, we developed MathFlowChart: an interactive HTML/SVG/React diagram mapping 221 nodes and 491 edges across modern mathematics. The chart is navigable (pan/zoom/click), color-coded by domain, and includes nodes for theorems, lemmas, and contested results.

The chart traces the path from categorical foundations through the Scholze program to TQFT:

- **Foundations:** Category Theory to Adjunctions, Monoidal Categories, Kan Extensions, Monads, Enriched Categories, Operads
- **Sheaf theory:** Topos Theory to Sheaf Cohomology to Etale Cohomology (the Grothendieck program)
- **Condensed math:** Condensed Sets to Condensed Abelian Groups to Liquid Vector Spaces, with the Scholze program (14 nodes: perfectoid spaces through motivic geometrization)
- **TQFT:** Algebraic Topology to Homotopy Theory to TQFT to Chern-Simons Theory, with connections to Gauge Theory, Conformal Field Theory, and String Theory

The chart also includes the Langlands program connections (geometric Langlands uses condensed infrastructure via Fargues-Fontaine), chromatic homotopy theory (which connects unexpectedly to modular forms via topological modular forms), and concatenation theory (the free monoid, connecting to logic, computation, linguistics, and HoTT via path composition).

The chart is distributed as a self-contained `index.html` file using React 18 and Babel standalone loaded from CDN, openable directly in any browser.

5. Roadmap for Further Development

5.1 Near-term (days to weeks)

Preserve finite limits for the embedding. The proof strategy is documented: factor through `sheafToPresheaf`, reduce to pointwise verification via `preservesLimit_of_evaluation`, use `ContinuousMap.piEquiv` for products and kernel factorization for equalizers, lift through the forget functor.

Complete the `Cob2` functor. Case analysis showing the functor action respects all Frobenius relations on the quotient morphism type.

Project integration. The five active files need proper import structure and deduplication.

5.2 Medium-term (weeks to months)

NormedSpaceCat and fully faithful embedding. Define the category of normed spaces over a nontrivially normed field R . Restrict the embedding and prove fullness (which holds in this setting, unlike for bare seminormed groups).

Concrete Frobenius object in CondensedAb. Construct an explicit commutative Frobenius algebra (e.g., condensed integers with multiplication and diagonal) to instantiate the first fully concrete 2d TQFT in the formalization.

Right exactness (cokernel preservation). Uses the open mapping theorem, specifically a Banach space result.

5.3 Long-term (months to years)

Projective tensor product of normed spaces. Definition, completion, universal property. Not in Mathlib. Would give MonoidalCategory on normed space categories and enable the monoidal embedding.

Port LTE exactness to current Mathlib. The liquid tensor product's exactness, proved in 2022 in a now-archived repository.

Smooth cobordism categories. Manifolds with corners, diffeomorphism classes, gluing. Unformalized in any proof assistant.

Concrete Chern-Simons / factorization algebra constructions valued in condensed abelian groups.

6. Methodology

6.1 The Iterative Process

The work was conducted in six sessions of AI-assisted formal verification. Each session consisted of:

1. A carefully scoped prompt from Claude to Aristotle, based on analysis of previous results
2. Aristotle producing a Lean 4 file with Mathlib API audit, constructions, and proofs
3. Claude analyzing the output, identifying gaps, and designing the next prompt

The prompts evolved from exploratory ("can you formalize this idea?") to targeted ("check whether this specific Mathlib module contains what we need"). The most productive prompts were those that front-loaded reconnaissance (searching Mathlib for existing infrastructure) before attempting construction. The Session 3 breakthrough, where the entire monoidal structure turned out to already exist, directly resulted from this "search before build" approach.

6.2 Tools

- **Lean 4** with **Mathlib v4.28.0** for formal verification
- **Harmonic Aristotle** for proof search, Mathlib API discovery, and counterexample

construction

- **Claude (Anthropic)** for mathematical synthesis, prompt engineering, analysis, and documentation
- **MathFlowChart** (React/SVG) for dependency visualization
- All formal results independently verifiable via `lake build`

7. Conclusion

Six rounds of AI-assisted formal verification, starting from a blank slate, produced approximately 1280 active lines of Lean 4 across 5 files with 3 sorries and 0 custom axioms. The central results: CondensedAb is a sorry-free symmetric monoidal abelian category, the TQFT transfer theorem is sorry-free, the embedding from seminormed groups exists sorry-free with the sheaf condition verified, and the commutative Frobenius algebra structure classifying 2d TQFTs is sorry-free.

The most significant finding was the disproof of fullness for the embedding SemiNormedGrp to CondensedAb. This identifies a precise mathematical boundary: condensed abelian groups see more morphisms between normed groups than the normed category itself does. Fullness requires restricting to normed vector spaces over a valued field. This was not anticipated and was discovered through the formalization process.

The remaining barriers are increasingly specific: projective tensor products in Mathlib (for the monoidal embedding), porting the Liquid Tensor Experiment’s exactness result (for full gluing), and smooth cobordism formalization (for geometric TQFTs). The most promising near-term mathematical application is reformulating Costello-Gwilliam [CostelloGwilliam2017] factorization algebras in the condensed setting.

The accompanying Lean 4 project (LiquidTQFT-project.zip) contains all formalized results and is buildable via `lake exe cache get && lake build` against Mathlib v4.28.0.

Appendix A: File Inventory

File	Lines	Sorries	Key Content
LiquidTQFT.lean	288	0	Abstract TQFT, transfer, monoidal CondensedAb
MonoidalViaLocalization.lean	227	0	Symmetric monoidal via localization
CondensedMonoidal.lean	567	5	Manual construction (superseded)

BanachEmbedding.lean	526	2	Embedding, faithfulness, fullness disproof
Cob2.lean	374	1	Frobenius algebras, 2d cobordisms
Active total	~1415	3	Excluding superseded CondensedMonoidal

Appendix B: Key Mathlib Dependencies

Module	Role
CategoryTheory.Sites.Monoidal	Sheaf.monoidalCategory, W.monoidal
CategoryTheory.Sites.Localization	presheafToSheaf is IsLocalization J.W
CategoryTheory.Localization.Monoidal.Basic	MonoidalCategory on localized categories
CategoryTheory.Monoidal.FunctorCategory	Pointwise monoidal on presheaf categories
Algebra.Category.ModuleCat.Monoidal	MonoidalCategory, MonoidalClosed on ModuleCat
Condensed.TopComparison	topCatToCondensedSet, yonedaPresheaf
Analysis.Normed.Group.SemiNormedGrp	SemiNormedGrp category
Topology.ContinuousMap.Compact	SeminormedAddCommGroup C(S,V)
CategoryTheory.Monoidal.Mon_	Mon_, Comon_ (monoid/comonoid objects)

Appendix C: Accompanying Materials

- **LiquidTQFT-project.zip**: Complete Lean 4 project, buildable via `lake exe cache get && lake build`
- **MathFlowChart (index.html)**: Interactive 221-node dependency map of modern mathematics
- **liquid_tqft_proposal.md**: Research proposal with status tables and gap analysis

Citation

Co-authored-by: Aristotle (Harmonic) <aristotle-harmonic@harmonic.fun>

Synthesis, prompt engineering, and mathematical analysis by Claude (Anthropic).

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